

Autodesk® Scaleform®

Scaleform 4

Scaleform (Scaleform 3) Scaleform 4.0

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Autodesk® Scaleform® 4.3

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Autodesk Scaleform

Scaleform 4

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1 Scaleform 4

Scaleform v4 Autodesk SDK UI
 SDK , ActionScript 3.0 ,Scaleform ,Scaleform v4
 AS 2.0 AS 3.0

AS3, SDK ,Scaleform v4

- w/ 2x+ – 2.5D
- - Scaleform 4 Advance Display
- – ,AS3 AS2 Scaleform 4
- - AS3 (Code Base), ,
- / – ,AS3
- – AS 2.0 , AS 3.0
 (Runtime Exception)
- – AS 2.0 , AS 3.0 ,
- – ActionScript 3.0 ,
 (Method Closure)

Scaleform 4 , Scaleform 4
 Scaleform 4.0 4.2 ,Scaleform 4 Scaleform 3.x
 , Scaleform
 Scaleform 4 , Scaleform C++
 : (NameSpace) ,

2

2.1

```
namespace  
{  
    class Foo  
    {  
        ...  
    };  
}
```

```
namespace Scaleform  
{  
    class Foo  
    {  
        ...  
    };  
}
```

- 1) `Scaleform::Foo var;`
- 2) `'using'`
`using namespace Scaleform;`
`Foo var;`

C++

Scaleform, "Scaleform"

- Scaleform, "using"

- `Scaleform` `"SF"` `(` `SF = Scaleform)`
- `(Header)` `;` `,` `,` `,`
- `SF` `,` `:"Render"` `"Sound"` `"GfX"`
- `,` `,"Win32"` `"PS3"` `,` `"Scaleform::Render::D3D9"` `,` `"SF::D3D9::Render"`

2.2 Scaleform

`Scaleform` `(Public)` `Scaleform`

`Scaleform` `Scaleform` `,` `,` `,`

`Scaleform 4`

- `Scaleform`
 - o `GfX`
 - `AMP`
 - `XML`
 - o `Render`
 - `Math2D`
 - `D3D9`
 - `GL`
 - `PS3`
 - `...`
 - o `Text`
 - o `Sound`
 - o `Alg`
 - o `UTF8Util`

3

Scaleform , Scaleform

, , ,

- "Scaleform" Src
- "Src/Kernel"

- Src
 - o GFx
 - AMP
 - AS2
 - AS3
 - ...
 - o Kernel
 - HeapMH
 - HeapPT
 - o Render
 - PS3
 - GL
 - D3D9
 - ...
 - o Sound
 - o XML

Src , cpp (Public) ,

4 Includes

```
Scaleform 3.x      ,Scaleform 4      /Src      ,      /Include
      .cpp

      ,      '      '(convenience)      ,Gfx.h
Scaleform      ;Gfx_Kernel.h      Kernel      ,      ,      Include

      ,
      ,
      include

#include "Kernel/SF_File.h"
#include "Gfx/Gfx_Player.h"
#include "Gfx/Gfx_Loader.h"
#include "Gfx/Gfx_Log.h"

#include "Gfx_Kernel.h"
#include "Gfx.h"

      ,      ,

"PublicHeader      :      GroupName"

,GroupName      ,      Gfx      Kernel      Render      AMP      ...      none      GroupName(      )      Gfx,
      Gfx.h      ,      GroupName      Kernel,      Gfx_Kernel.h      ,
GroupName      'none',      -      ,
,
```



```
loader.SetAS3Support(pASSupport);
```

```
, Scaleform 3.x Scaleform 4
GfxCompat.h
```

5.1

```
Scaleform 3.x 4 , , Scaleform 4
```

```
#include "Gfx_Kernel.h"
#include "Gfx.h"
#include "Gfx_Renderer_D3D9.h" // or whatever platform you need
```

```
(Gfx.h) AS3 AS3
"Gfx/AS3/AS3_Global.h" AS3
Scaleform LITE
```

AS3_Global.h and Obj\AS3_Obj_Global.xxx

```
:AS3_Global.h Obj\AS3_Obj_global.xxx
AS3_Obj_Global.xxx " "ActionScript 3 swf "
"(script) , GlobalObjectCPP, C++
Scaleform VM

AS3_Global.h ClassRegistrationTable
AS3 C++ , ,
ClassRegistrationTable, ,
AS3_Global.h

ClassRegistrationTable include
ClassRegistrationTable( ) AS3_Global.h
, ( , ),
, AS3 VM
, " " ,
```

5.2

, GFC SF_, Scaleform GFX_

Scaleform 3.X	Scaleform 4
GFC_FX_VERSION_STRING	GFX_VERSION_STRING
GUNUSED	SF_UNUSED
GFC_64BIT_POINTERS	SF_64BIT_POINTERS
...	...

5.3 Namespace

Scaleform 4 , (Scaleform::GFx::Event) 'using namespace' , using cpp ,

```
namespace SF = Scaleform;
using namespace Scaleform;
using namespace Render;
using namespace GFx;
```

5.4

Scaleform 4 , ,

Scaleform 3.X	Scaleform 4
UInt	unsigned
SInt	int
Float	float
...	

Scaleform 4 Scaleform , ,
 Scaleform 3.x , GFxEvent , Scaleform 4 , GFx::Event
 Scaleform 3.x Scaleform 4

Scaleform 3.X	Scaleform 4	Scaleform 4
		'using namespace ...'
GFxSystem	Scaleform::GFx::System	System

GPtr	Scaleform::Ptr	Ptr
GFxMovieDef	Scaleform::GFx::MovieDef	MovieDef
GFxMovieView	Scaleform::GFx::Movie	Movie
GRendererD3D9	Scaleform::Render::D3D9::Renderer	Render::D3D9::Renderer
GFxRenderConfig	Scaleform::GFx::RenderConfig	RenderConfig
GString	Scaleform::String	String
GFxFSCCommandHandler	Scaleform::GFx::FSCCommandHandler	FSCCommandHandler
GFxLog	Scaleform::GFx::Log	Log
GFxImageCreator	Scaleform::GFx::ImageCreator	ImageCreator
GFxKey	Scaleform::GFx::Key	Key
GFxKeyEvent	Scaleform::GFx::KeyEvent	KeyEvent
GFxCharEvent	Scaleform::GFx::CharEvent	CharEvent
GFxEvent	Scaleform::GFx::Event	Event
GMatrix2D	Scaleform::Render::Matrix2x4	Matrix2x4
GMatrix3D	Scaleform::Render::Matrix3x4	Matrix3x4
GMatrix3D	Scaleform::Render::Matrix4x4	Matrix4x4
GRect	Scaleform::Render::Rect	Rect
...	...	...

6 Scaleform 4.0 4.2

Scaleform 4.2 4.0 API Scaleform 4.1
API 4.0

6.1

SF 4.2 PCRE Perl (Perl Compatible Regular
Expressions, PCRE), AS3 3rdParty , Scaleform 4.2
PCRE AS3, PCRE(
Include/GFxConfig.h SF_ENABLE_PCRE), PCRE

6.2

Scaleform 4.2 , , Scaleform Scaleform
(Build System)
Render() Sound() SF 4.2

Render Scaleform
X360/D3D9/D3D1x , InitHAL()
• HALConfig_DynamicShaderCompile

Visual Studio , : Render , ShaderData.xml,
Custom Build Step

6.3

Scaleform 4.2 -Scaleform 4.2
40 SF 4.0 Scaleform 4.0

gfxlicense.txt sf_license.txt
gfx.lib
,sf_license.txt Scaleform AMP Scaleform

Scaleform IME sf_video_license.txt
sf_ime_license.txt

Scaleform